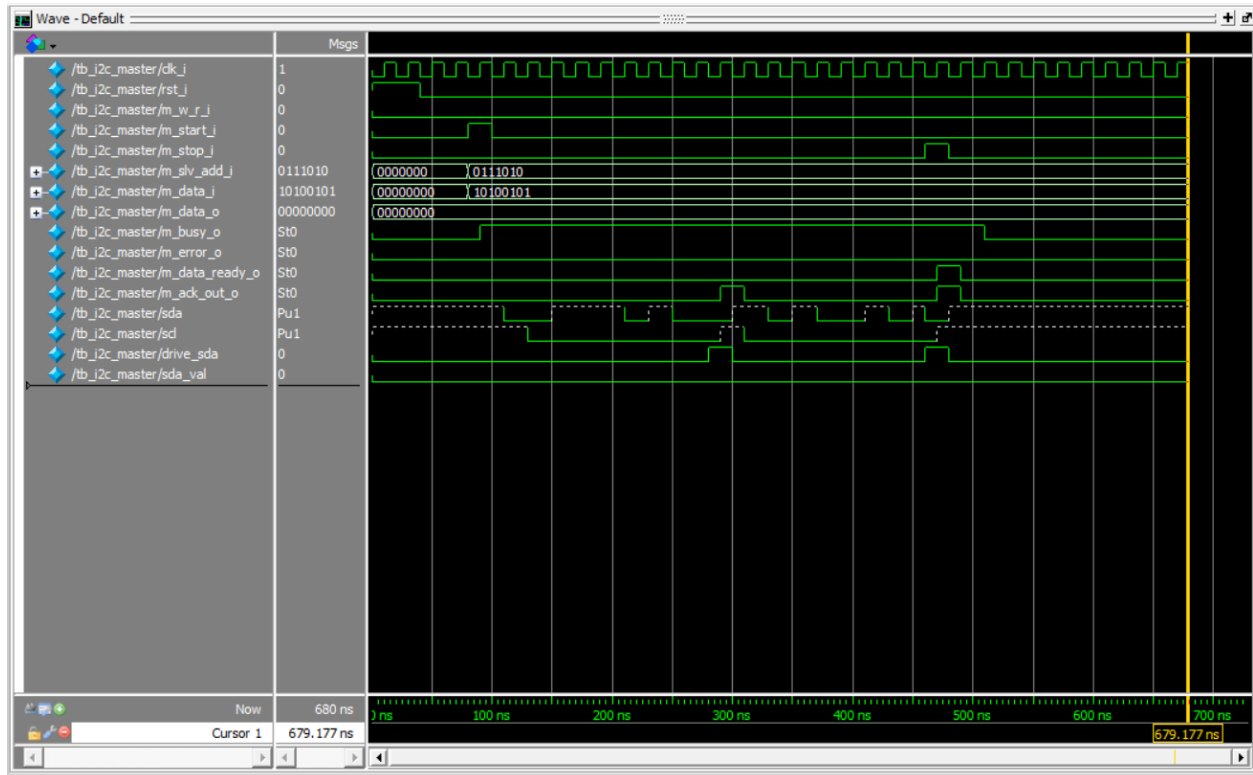


I2C Simulation Documentation



- **Reset Sequence:**

Initially, rst_i is asserted HIGH to transition the master into its reset state and clear any residual data. During this state, all control signals remain

- **START Condition:**

A pulse is driven on m_start_i, asserting m_busy_o HIGH. While scl stays in high-impedance, sda transitions to LOW to signal the start of transmission.

- **Address Phase:**

The sda line transmits the target slave address (0111010) followed by the write command bit (0). m_ack_out_o is asserted HIGH as an acknowledgment that the slave received the

address and is ready. During this phase, `drive_sda` is asserted, indicating that bus control has transferred from the master to the slave.

- **Data Transmission:**

Control shifts back as `drive_sda` deasserts, and the master transmits its data byte (10100101) over `sda`.

- **Data Acknowledgment:**

Upon reception, the slave takes control, `m_data_ready_o` is asserted HIGH, and the slave sends the acknowledgment via `m_ack_out_o`.

- **STOP Condition & Bus Idle:**

The `m_busy_o` signal drops back to LOW, `m_stop_i` is asserted, and both `sda` and `scl` return to their default high-impedance states.